Matemáticas
1º ESO

SOLUCIONES DE LA FICHA · UNIDAD 4

Fracciones

FentMates

Intenta resolver cada ejercicio antes de mirar su solución. Si no te sale el mismo resultado, repasa los pasos o consulta los apuntes interactivos de la unidad. Recuerda que los resultados se tienen que dar siempre en forma de fracción irreducible.

1 El concepto de fracción

1. ★☆☆

a) $\frac{4}{10}$ **b)** $\frac{3}{4}$ **c)** $\frac{1}{7}$ **d)** $\frac{2}{4}$ **e)** $\frac{5}{7}$ **f)** $\frac{4}{6}$

2. ★☆☆

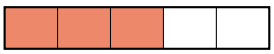
a) num. 3, den. 5: tres quintos

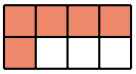
b) num. 4, den. 6: cuatro sextos

c) num. 12, den. 15: doce quinceavos

d) num. 4, den. 3: cuatro tercios

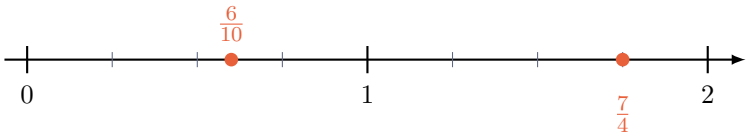
3. ★☆☆

a) $\frac{3}{5}$ 

b) $\frac{5}{8}$ 

c) $\frac{2}{3}$ 

4. ★☆☆ Dividimos cada unidad en 10 partes para $\frac{6}{10}$ y en 4 para $\frac{7}{4} = 1 + \frac{3}{4}$:



5. ★★☆

a) Impropia: $2 + \frac{1}{3}$

b) Propia

c) Igual a 1

d) Impropia: $3 + \frac{1}{4}$

e) Propia

f) Impropia: $3 + \frac{2}{6}$

6. ★★☆

a) $\frac{15}{60}$ **b)** $\frac{4}{12}$ **c)** $\frac{5}{7}$ **d)** $\frac{300}{1000}$

7. ★★★ Entre $\frac{9}{9}$ y $\frac{18}{9}$: $\boxed{\frac{10}{9}, \frac{11}{9}, \frac{12}{9}, \frac{13}{9}, \frac{14}{9}, \frac{15}{9}, \frac{16}{9}, \frac{17}{9}}$. Entre $\frac{12}{6}$ y $\frac{18}{6}$: $\boxed{\frac{13}{6}, \frac{14}{6}, \frac{15}{6}, \frac{16}{6}, \frac{17}{6}}$.

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2 Fracción de un número

8. ★☆☆

a) $(48 : 6) \cdot 5 = \boxed{40}$ **b)** $(490 : 7) \cdot 4 = \boxed{280}$ **c)** $(200 : 5) \cdot 3 = \boxed{120}$

d) $(32 : 8) \cdot 7 = \boxed{28}$ **e)** $(60 : 5) \cdot 2 = \boxed{24}$ **f)** $(180 : 6) \cdot 3 = \boxed{90}$

9. ★☆☆ Chicos: $(30 : 5) \cdot 2 = \boxed{12}$. Chicas: $30 - 12 = \boxed{18}$.

10. ★★☆ $(2400 : 8) \cdot 3 = \boxed{900\ \text{L}}$. Faltan $2400 - 900 = \boxed{1500\ \text{L}}$.

11. ★★☆

a) $36 : 3 = 12$; $12 \cdot 4 = \boxed{48}$ **b)** $18 : 2 = 9$; $9 \cdot 7 = \boxed{63}$

12. ★★☆ Le quedan $\frac{2}{5}$ de la paga, que son 14 €: $14 : 2 = 7$; $7 \cdot 5 = \boxed{35\ €}$.

13. ★★★ $\frac{3}{4}$ de $360 = 270$; $\frac{2}{3}$ de $270 = \boxed{180}$.

14. ★★★ 1º: $(84 : 7) \cdot 3 = \boxed{36}$. 2º: $84 : 4 = \boxed{21}$. 3º: $84 - 36 - 21 = \boxed{27}$.

3 Porcentajes y fracciones

15. ★☆☆

a) $\frac{20 \cdot 180}{100} = \boxed{36}$ **b)** $\frac{45 \cdot 80}{100} = \boxed{36}$ **c)** $\frac{12 \cdot 350}{100} = \boxed{42}$

d) $\frac{85 \cdot 1300}{100} = \boxed{1105}$ **e)** $\frac{50 \cdot 64}{100} = \boxed{32}$ **f)** $\frac{25 \cdot 120}{100} = \boxed{30}$

16. ★☆☆

a) $\frac{50}{100} = \frac{1}{2}$ **b)** $\frac{25}{100} = \frac{1}{4}$ **c)** $\frac{75}{100} = \frac{3}{4}$ **d)** $\frac{20}{100} = \frac{1}{5}$

e) $\frac{10}{100} = \frac{1}{10}$ **f)** $\frac{40}{100} = \frac{2}{5}$ **g)** $\frac{5}{100} = \frac{1}{20}$ **h)** $\frac{35}{100} = \frac{7}{20}$

17. ★☆☆

a) $\frac{1}{2} = \frac{50}{100} = \boxed{50\,\%}$ **b)** $\frac{3}{4} = \frac{75}{100} = \boxed{75\,\%}$ **c)** $\frac{2}{5} = \frac{40}{100} = \boxed{40\,\%}$

d) $\frac{7}{10} = \frac{70}{100} = \boxed{70\,\%}$ **e)** $\frac{9}{20} = \frac{45}{100} = \boxed{45\,\%}$ **f)** $\frac{3}{25} = \frac{12}{100} = \boxed{12\,\%}$

18. ★★☆ $25\,\% = \frac{1}{4}$: descuento $80 : 4 = \boxed{20\ €}$; pagamos $80 - 20 = \boxed{60\ €}$.

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19. ★★☆ Chicas: $\frac{52 \cdot 650}{100} = \boxed{338}$. Chicos: $650 - 338 = \boxed{312}$.

20. ★★★ De aquí: $100 - 7 - 25 - 12 = \boxed{56\,\%}$. Personas: $\boxed{26\,460}$, $\boxed{94\,500}$, $\boxed{45\,360}$ y $\boxed{211\,680}$.

21. ★★★ Han aprobado 17: $\frac{17}{25} = \frac{68}{100} = \boxed{68\,\%}$.

4 Fracciones equivalentes

22. ★☆☆

a) $\frac{1}{4} = \frac{2}{8} = \frac{3}{12} = \frac{4}{16} = \frac{5}{20}$

b) $\frac{2}{5} = \frac{4}{10} = \frac{6}{15} = \frac{8}{20} = \frac{10}{25}$

c) $\frac{5}{15} = \frac{10}{30} = \frac{15}{45} = \frac{20}{60} = \frac{25}{75}$

d) $\frac{7}{13} = \frac{14}{26} = \frac{21}{39} = \frac{28}{52} = \frac{35}{65}$

e) $\frac{11}{17} = \frac{22}{34} = \frac{33}{51} = \frac{44}{68} = \frac{55}{85}$

f) $\frac{1}{10} = \frac{2}{20} = \frac{3}{30} = \frac{4}{40} = \frac{5}{50}$

23. ★☆☆

a) $\frac{4}{12} = \frac{2}{6} = \frac{1}{3}$

b) $\frac{12}{20} = \frac{6}{10} = \frac{3}{5}$

c) $\frac{66}{84} = \frac{33}{42} = \frac{11}{14}$

d) $\frac{15}{45} = \frac{5}{15} = \frac{1}{3}$

e) $\frac{50}{100} = \frac{25}{50} = \frac{1}{2}$

(Hay otras respuestas posibles.)

24. ★☆☆

a) No: $\frac{5}{7}$

b) Sí: $\text{m.c.d.}(5,7) = 1$

c) No: $\frac{2}{1} = 2$

25. ★★☆

a) $\frac{4}{6} = \boxed{\frac{2}{3}}$

b) $\frac{4}{20} = \boxed{\frac{1}{5}}$

c) $\frac{3}{12} = \boxed{\frac{1}{4}}$

d) $\frac{2}{6} = \boxed{\frac{1}{3}}$

e) $\frac{35}{40} = \boxed{\frac{7}{8}}$

f) $\frac{75}{100} = \boxed{\frac{3}{4}}$

g) $\frac{4}{16} = \boxed{\frac{1}{4}}$

h) $\frac{14}{21} = \boxed{\frac{2}{3}}$

i) $\frac{10}{15} = \boxed{\frac{2}{3}}$

j) $\frac{25}{45} = \boxed{\frac{5}{9}}$

k) $\frac{13}{52} = \boxed{\frac{1}{4}}$

l) $\frac{33}{34} = \boxed{\frac{33}{34}}$

m) $\frac{705}{120} = \boxed{\frac{47}{8}}$

n) $\frac{170}{105} = \boxed{\frac{34}{21}}$

ñ) $\frac{256}{512} = \boxed{\frac{1}{2}}$

26. ★★☆

a) $\frac{54}{24} = \frac{9}{4}$

b) $\frac{40}{88} = \frac{5}{11}$

c) $\frac{81}{36} = \frac{9}{4}$

d) $\frac{72}{32} = \frac{9}{4}$

e) $\frac{75}{185} = \frac{15}{37}$

f) $\frac{50}{110} = \frac{5}{11}$

Equivalentes: $\frac{54}{24}$, $\frac{81}{36}$ y $\frac{72}{32}$ (todas valen $\frac{9}{4}$); $\frac{40}{88}$ y $\frac{50}{110}$ (valen $\frac{5}{11}$).

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27. ★★★ $\frac{15}{24} = \frac{5}{8} = \boxed{\frac{10}{16}}$ $\quad \frac{5}{8} = \frac{5 \cdot 16}{8 \cdot 16} = \boxed{\frac{80}{128}}$

28. ★★★ $\boxed{\frac{14}{21}, \frac{16}{24}, \frac{18}{27}, \frac{20}{30}}$

5 Comprobar si dos fracciones son equivalentes

29. ★☆☆

a) $8 \cdot 12 = 96$, $32 \cdot 3 = 96$: $\boxed{\text{sí}}$

b) $6 \cdot 34 = 204$, $15 \cdot 26 = 390$: $\boxed{\text{no}}$

c) $7 \cdot 46 = 322$, $23 \cdot 14 = 322$: $\boxed{\text{sí}}$

d) $1 \cdot 30 = 30$, $2 \cdot 15 = 30$: $\boxed{\text{sí}}$

e) $68 \cdot 13 = 884$, $52 \cdot 17 = 884$: $\boxed{\text{sí}}$

f) $3 \cdot 67 = 201$, $29 \cdot 33 = 957$: $\boxed{\text{no}}$

30. ★☆☆

a) $x = \boxed{9}$ **b)** $x = \boxed{32}$ **c)** $x = \boxed{3}$ **d)** $x = \boxed{98}$ **e)** $x = \boxed{33}$

31. ★★☆ $\frac{2}{5} = \frac{\boxed{8}}{20} = \frac{12}{\boxed{30}} = \frac{\boxed{18}}{45}$

32. ★★☆ $\boxed{\frac{14}{20}}$ $(= \frac{7}{10})$; las demás valen $\frac{2}{3}$.

33. ★★★ $\frac{15}{20} = \frac{3}{4}$: $x = \boxed{9}$, $y = \boxed{12}$.

34. ★★★ En $\frac{3}{5}$ la diferencia es 2; amplificamos por $24 : 2 = 12$: $\boxed{\frac{36}{60}}$.

6 Reducción a común denominador

35. ★☆☆

a) $\frac{7}{35}, \frac{15}{35}$

b) $\frac{4}{18}, \frac{3}{18}$

c) $\frac{6}{8}, \frac{3}{8}$

d) $\frac{8}{28}, \frac{7}{28}$

e) $\frac{10}{42}, \frac{63}{42}$

f) $\frac{6}{10}, \frac{1}{10}$

g) $\frac{3}{12}, \frac{2}{12}$

h) $\frac{3}{7}, \frac{35}{7}$

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36. ★★☆

a) m.c.m. $= 48$: $\frac{14}{48}, \frac{9}{48}$

b) m.c.m. $= 24$: $\frac{12}{24}, \frac{16}{24}, \frac{15}{24}$

c) m.c.m. $= 30$: $\frac{105}{30}, \frac{12}{30}, \frac{9}{30}$

d) m.c.m. $= 24$: $\frac{15}{24}, \frac{12}{24}, \frac{14}{24}$

e) m.c.m. $= 42$: $\frac{24}{42}, \frac{14}{42}, \frac{15}{42}$

f) m.c.m. $= 180$: $\frac{30}{180}, \frac{24}{180}, \frac{35}{180}$

g) m.c.m. $= 28$: $\frac{24}{28}, \frac{4}{28}, \frac{12}{28}$

h) m.c.m. $= 60$: $\frac{100}{60}, \frac{25}{60}, \frac{45}{60}, \frac{27}{60}$

37. ★★☆ $\boxed{\frac{30}{36}, \frac{21}{36}, \frac{16}{36}}$

38. ★★★ $\frac{2}{3}, \frac{2}{3}, \frac{3}{4} \rightarrow$ m.c.m.$(3,3,4) = 12$: $\boxed{\frac{8}{12}, \frac{8}{12}, \frac{9}{12}}$.

7 Comparar y ordenar fracciones

39. ★☆☆

a) $\frac{4}{7} < \frac{5}{7} < \frac{6}{7}$

b) $\frac{3}{9} < \frac{4}{9} < \frac{5}{9}$

c) $\frac{2}{13} < \frac{3}{13} < \frac{4}{13}$

(Hay muchas respuestas posibles.)

40. ★☆☆

a) $\frac{9}{5} \boxed{>} \frac{9}{8}$

b) $\frac{6}{11} \boxed{<} \frac{10}{11}$

c) $\frac{3}{7} \boxed{<} \frac{3}{4}$

d) $\frac{4}{8} \boxed{>} \frac{2}{16}$

e) $\frac{2}{5} \boxed{<} \frac{5}{6}$

f) $\frac{1}{3} \boxed{<} \frac{3}{7}$

g) $\frac{8}{9} \boxed{<} \frac{11}{12}$

h) $\frac{3}{4} \boxed{=} \frac{24}{32}$

41. ★★☆

a) $\frac{2}{7} < \frac{3}{5} < \frac{11}{2}$

b) $\frac{4}{9} < \frac{2}{3} < \frac{5}{6}$

c) $\frac{1}{10} < \frac{2}{6} < \frac{5}{4}$

d) $\frac{23}{21} < \frac{14}{7} < \frac{18}{3}$

e) $\frac{3}{24} < \frac{2}{12} < \frac{4}{15}$

f) $\frac{3}{28} < \frac{12}{14} = \frac{6}{7}$

42. ★★☆

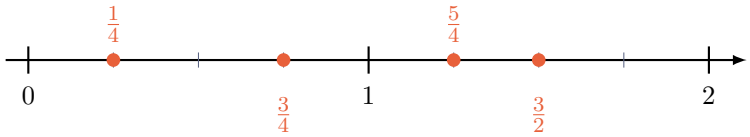
a) $\frac{2}{8} < \frac{3}{10} < \frac{7}{20} < \frac{4}{5}$

b) $\frac{4}{7} < \frac{2}{3} < \frac{5}{4} < \frac{5}{2}$

c) $\frac{1}{4} < \frac{2}{6} < \frac{7}{12} < \frac{3}{2}$

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43. ★★☆



44. ★★★ $\frac{2}{5}=\frac{4}{10}$ y $\frac{3}{5}=\frac{6}{10}$: $\boxed{\frac{5}{10}}$. $\frac{1}{3}=\frac{4}{12}$ y $\frac{1}{2}=\frac{6}{12}$: $\boxed{\frac{5}{12}}$.

45. ★★★ m.c.m. $=40$: $\frac{30}{40}, \frac{25}{40}, \frac{28}{40}$. Ha leído más $\boxed{\text{Ana}}$, después Clara y después Gabriel.

8 Sumas y restas

46. ★☆☆

a) $\frac{4}{7}+\frac{8}{7}=\boxed{\frac{12}{7}}$ **b)** $\frac{2}{3}+\frac{1}{5}=\boxed{\frac{13}{15}}$ **c)** $\frac{5}{2}-\frac{1}{8}=\boxed{\frac{19}{8}}$

d) $\frac{3}{6}-\frac{4}{12}=\boxed{\frac{1}{6}}$ **e)** $\frac{5}{8}+\frac{1}{5}=\boxed{\frac{33}{40}}$ **f)** $\frac{14}{15}-\frac{2}{3}=\boxed{\frac{4}{15}}$

47. ★☆☆

a) $\frac{3}{4}+\frac{6}{8}=\boxed{\frac{3}{2}}$ **b)** $4+\frac{25}{9}=\boxed{\frac{61}{9}}$ **c)** $\frac{25}{8}-2=\boxed{\frac{9}{8}}$

d) $\frac{3}{5}+1=\boxed{\frac{8}{5}}$ **e)** $5-\frac{3}{25}=\boxed{\frac{122}{25}}$ **f)** $\frac{13}{6}+\frac{2}{7}=\boxed{\frac{103}{42}}$

48. ★★☆

a) $\frac{12}{15}+\frac{8}{9}=\boxed{\frac{76}{45}}$ **b)** $\frac{7}{12}-\frac{3}{10}=\boxed{\frac{17}{60}}$ **c)** $\frac{7}{6}+\frac{8}{12}=\boxed{\frac{11}{6}}$

d) $\frac{18}{5}-\frac{3}{10}=\boxed{\frac{33}{10}}$ **e)** $\frac{5}{3}-\frac{9}{24}=\boxed{\frac{31}{24}}$ **f)** $\frac{35}{36}+\frac{2}{9}=\boxed{\frac{43}{36}}$

49. ★★☆

a) $\frac{4}{5}+\frac{2}{7}-\frac{1}{2}=\boxed{\frac{41}{70}}$ **b)** $\frac{6}{8}+\frac{4}{3}-\frac{8}{12}=\boxed{\frac{17}{12}}$

c) $\frac{3}{2}+\frac{6}{7}-\frac{2}{4}=\boxed{\frac{13}{7}}$ **d)** $6-\frac{1}{2}-\frac{4}{12}=\boxed{\frac{31}{6}}$

e) $\frac{2}{3}+\frac{1}{5}-\frac{1}{6}=\boxed{\frac{7}{10}}$ **f)** $\frac{4}{3}+\frac{1}{6}-\frac{2}{7}=\boxed{\frac{17}{14}}$

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50. ★★★

a) $\frac{2}{10}+\frac{5}{60}-\frac{1}{30}=\boxed{\frac{1}{4}}$

b) $\frac{12}{5}+2-\frac{1}{25}=\boxed{\frac{109}{25}}$

c) $\frac{6}{9}+\frac{1}{3}-\frac{2}{6}=\boxed{\frac{2}{3}}$

d) $\frac{10}{3}-\frac{2}{6}-\frac{4}{5}+\frac{1}{2}=\boxed{\frac{27}{10}}$

e) $4-\frac{1}{2}+\frac{1}{3}+2=\boxed{\frac{35}{6}}$

f) $\frac{7}{9}+\frac{1}{3}-\frac{2}{5}=\boxed{\frac{32}{45}}$

51. ★★★ $\frac{1}{4}+\frac{2}{5}+\frac{1}{6}=\frac{49}{60}$, falta $1-\frac{49}{60}=\boxed{\frac{11}{60}}$.

9 Multiplicación y división

52. ★☆☆

a) $\frac{6}{4}\cdot\frac{2}{5}=\boxed{\frac{3}{5}}$

b) $\frac{3}{2}\cdot\frac{2}{3}=\boxed{1}$

c) $2\cdot\frac{5}{6}=\boxed{\frac{5}{3}}$

d) $\frac{3}{2}\cdot\frac{4}{7}\cdot 3=\boxed{\frac{18}{7}}$

e) $\frac{3}{4}\cdot\frac{10}{2}\cdot\frac{1}{6}=\boxed{\frac{5}{8}}$

f) $\frac{4}{3}\cdot\frac{2}{8}\cdot\frac{6}{2}=\boxed{1}$

En el b) y el f) el producto es 1: las fracciones son **inversas**.

53. ★☆☆

a) $\boxed{\frac{2}{5}}$

b) $\boxed{\frac{1}{5}}$

c) $\boxed{4}$

d) $\boxed{\frac{3}{7}}$

54. ★☆☆

a) $\frac{2}{5}\cdot\frac{5}{3}=\frac{2}{3}$: $\boxed{\text{no}}$

b) $\frac{4}{3}\cdot\frac{12}{16}=1$: $\boxed{\text{sí}}$

c) $\frac{7}{9}\cdot\frac{9}{7}=1$: $\boxed{\text{sí}}$

d) $\frac{2}{4}\cdot\frac{4}{2}=1$: $\boxed{\text{sí}}$

e) $\frac{5}{6}\cdot\frac{36}{30}=1$: $\boxed{\text{sí}}$

f) $\frac{12}{13}\cdot\frac{26}{12}=2$: $\boxed{\text{no}}$

55. ★☆☆

a) $\frac{13}{4}$

b) $\frac{7}{3}$

c) $\frac{8}{15}$

d) $\frac{1}{6}$

56. ★★☆

a) $\frac{4}{3}:\frac{5}{6}=\boxed{\frac{8}{5}}$

b) $\frac{5}{4}:2=\boxed{\frac{5}{8}}$

c) $\frac{4}{8}:\frac{6}{3}=\boxed{\frac{1}{4}}$

d) $\frac{2}{6}:\frac{5}{4}:\frac{1}{3}=\boxed{\frac{4}{5}}$

e) $\frac{3}{4}:\frac{12}{5}:\frac{2}{5}=\boxed{\frac{25}{32}}$

f) $6:\frac{2}{3}:\frac{1}{5}=\boxed{45}$

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57. ★★☆

a) $\frac{3}{4}\cdot\frac{5}{8}:\frac{2}{5}=\boxed{\frac{75}{64}}$

b) $\frac{5}{4}:\frac{1}{3}\cdot\frac{6}{5}=\boxed{\frac{9}{2}}$

c) $\frac{5}{12}\cdot\frac{3}{2}:\frac{1}{6}=\boxed{\frac{15}{4}}$

d) $\frac{6}{5}:\frac{7}{4}\cdot\frac{2}{10}=\boxed{\frac{24}{175}}$

58. ★★★ $12:\frac{3}{4}=\frac{48}{3}=\boxed{16}$ botellas. $\frac{9}{10}:\frac{3}{4}=\frac{36}{30}=\boxed{\frac{6}{5}}$.

10 Operaciones combinadas

59. ★★☆

a) $\frac{5}{2}+\frac{1}{3}\cdot\frac{4}{5}=\boxed{\frac{83}{30}}$

b) $\frac{21}{4}-\frac{2}{5}:\frac{4}{3}=\boxed{\frac{99}{20}}$

c) $\frac{4}{3}:\frac{2}{6}+\frac{1}{3}\cdot\frac{5}{4}=\boxed{\frac{53}{12}}$

d) $\frac{4}{3}\cdot\left(\frac{2}{5}+\frac{1}{3}\right)=\boxed{\frac{44}{45}}$

e) $\left(\frac{2}{5}+1\right):\frac{8}{3}=\boxed{\frac{21}{40}}$

f) $\left(\frac{5}{4}+1\right)\cdot\frac{6}{7}=\boxed{\frac{27}{14}}$

60. ★★☆

a) $\left(\frac{2}{5}+\frac{3}{7}\right)\cdot\frac{1}{2}-\frac{4}{10}=\boxed{\frac{1}{70}}$

b) $\left(\frac{5}{4}+\frac{2}{3}+\frac{1}{12}\right)\cdot\frac{5}{4}-1=\boxed{\frac{3}{2}}$

c) $\frac{5}{4}:\left(\frac{6}{4}-1\right)+\frac{2}{5}=\boxed{\frac{29}{10}}$

d) $\frac{4}{3}-\left(\frac{1}{2}+\frac{2}{7}\right)+\frac{4}{5}\cdot\frac{2}{4}=\boxed{\frac{199}{210}}$

61. ★★★

a) $\left(\frac{1}{4}+\frac{2}{3}-\frac{1}{12}\right)\cdot 3+\left(\frac{3}{2}-\frac{5}{4}\right):\frac{1}{2}=\boxed{3}$

b) $4\cdot\left(\frac{2}{3}+\frac{4}{6}-1\right)+\frac{2}{3}=\boxed{2}$

c) $\left(\frac{3}{2}+\frac{1}{7}\right)\cdot\left(\frac{5}{3}-\frac{7}{15}\right)=\boxed{\frac{69}{35}}$

d) $\frac{12}{10}:\frac{5}{4}+\frac{1}{2}\cdot\frac{13}{5}=\boxed{\frac{113}{50}}$

e) $\left(\frac{4}{3}+\frac{2}{5}\cdot\frac{1}{3}\right):\frac{4}{12}=\boxed{\frac{22}{5}}$

f) $\frac{3}{2}:\frac{1}{4}\cdot\frac{4}{20}+\frac{3}{2}\cdot\frac{7}{5}=\boxed{\frac{33}{10}}$

62. ★★★

a) $\left(\frac{5}{4}+\frac{2}{3}-1\right)\cdot\frac{3}{5}+\left(\frac{2}{5}+\frac{4}{3}\right)\cdot 2=\boxed{\frac{241}{60}}$

b) $\frac{3}{8}+\frac{4}{5}\cdot\frac{1}{3}+\left(\frac{2}{6}-\frac{1}{4}\right)\cdot 3=\boxed{\frac{107}{120}}$

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11 Problemas con fracciones

63. ★☆☆

a) m.c.m. $= 280$: $\frac{40}{280} + \frac{112}{280} + \frac{35}{280} = \boxed{\frac{187}{280}}$

b) $1 - \frac{187}{280} = \boxed{\frac{93}{280}}$

c) Más: $\boxed{\text{Elena}}$ ($\frac{112}{280}$); menos: $\boxed{\text{Maite}}$ ($\frac{35}{280}$).

64. ★☆☆ Nos hemos comido $(24 : 8) \cdot 5 = 15$: quedan $\boxed{9}$ porciones.

65. ★★☆

a) $\frac{1}{2} + \frac{1}{7} + \frac{2}{7} = \frac{13}{14}$: sobra $\boxed{\frac{1}{14}}$.

b) $\frac{1}{14} + \frac{3}{5} = \frac{5}{70} + \frac{42}{70} = \boxed{\frac{47}{70}}$ de pizza.

66. ★★☆

a) $1 - \frac{1}{2} - \frac{1}{5} = \boxed{\frac{3}{10}}$

b) Más de 65: $\boxed{2625}$; menos de 18: $\boxed{1050}$; entre 18 y 65: $\boxed{1575}$.

67. ★★☆ Se gastan 30 L (quedan 30); después $\frac{1}{3}$ de $30 = 10$ L (quedan 20); se derrama la mitad: quedan $\boxed{10\text{ L}}$.

68. ★★★ $\frac{35}{105} + \frac{21}{105} + \frac{30}{105} = \frac{86}{105} \Rightarrow 1 - \frac{86}{105} = \boxed{\frac{19}{105}}$.

69. ★★★

a) Hipoteca $\boxed{1800\,€}$, comida $\boxed{1080\,€}$, coche $\boxed{600\,€}$; quedan $\boxed{1920\,€}$.

b) $\frac{1920}{5400} = \boxed{\frac{16}{45}}$

70. ★★★ Queda $\frac{1}{3}$ de $\frac{3}{4}$, es decir, $\frac{1}{4}$ del depósito, que son 50 L: capacidad $50 \cdot 4 = \boxed{200\text{ L}}$.

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